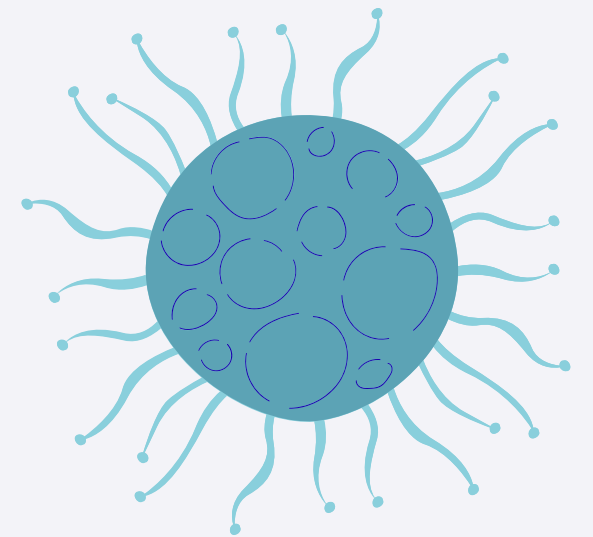
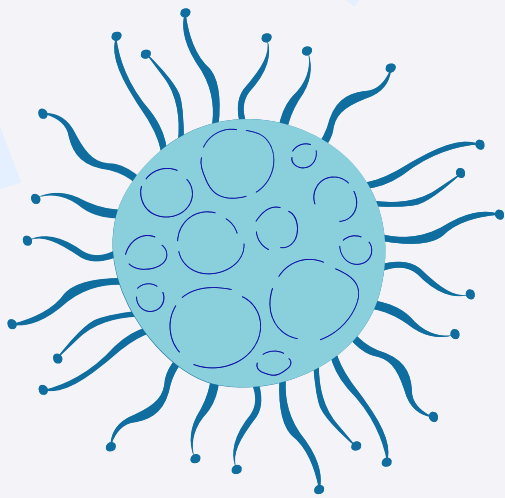
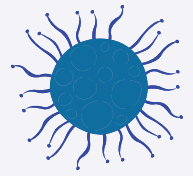




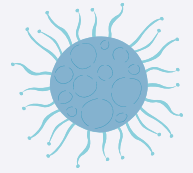
PROJECT 12: LONGITUDINAL BIOMARKER TRAJECTORY IN ONCOLOGY

By Joanne Mittapalli and Vaishnavi Dharmasoth

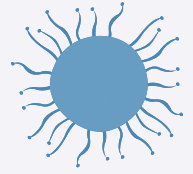




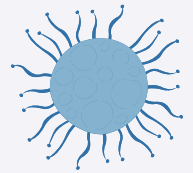
Why It Matters



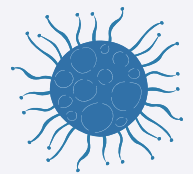
Dataset Overview



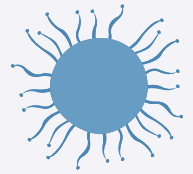
Three Models



Key Findings



Limitations



Takeaways

AGENDA

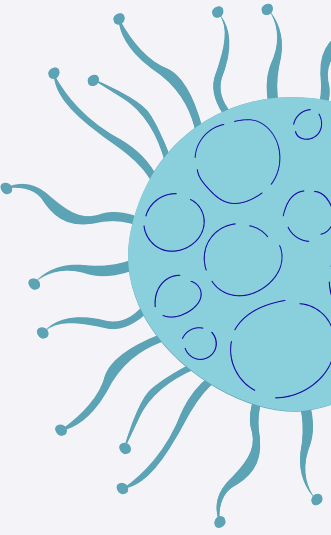
WHY IT MATTERS

CENSORED $\neq$ CURED

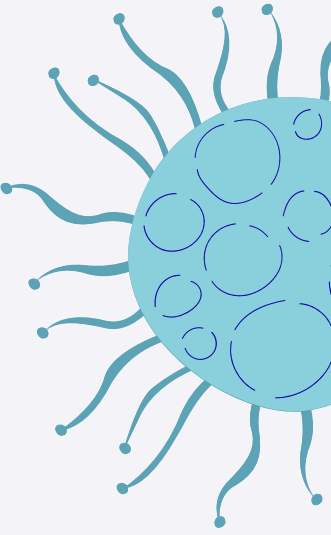
A patient whose PSA hasn't risen yet is NOT the same as a patient who will never recur. A clinician sees a post-treatment PSA value and asks: "Is this good news or just early?" We don't know whether to intensify treatment, switch therapy, or wait.

Solution: Use the entire PSA trajectory, not single measurements

The rate of PSA decline after treatment, nadir value, and velocity of rise at relapse carry more prognostic information than any isolated number. Together, survival analysis and biomarker trajectory modeling tell us which patients will truly recur and when.



PSA TRAJECTORY

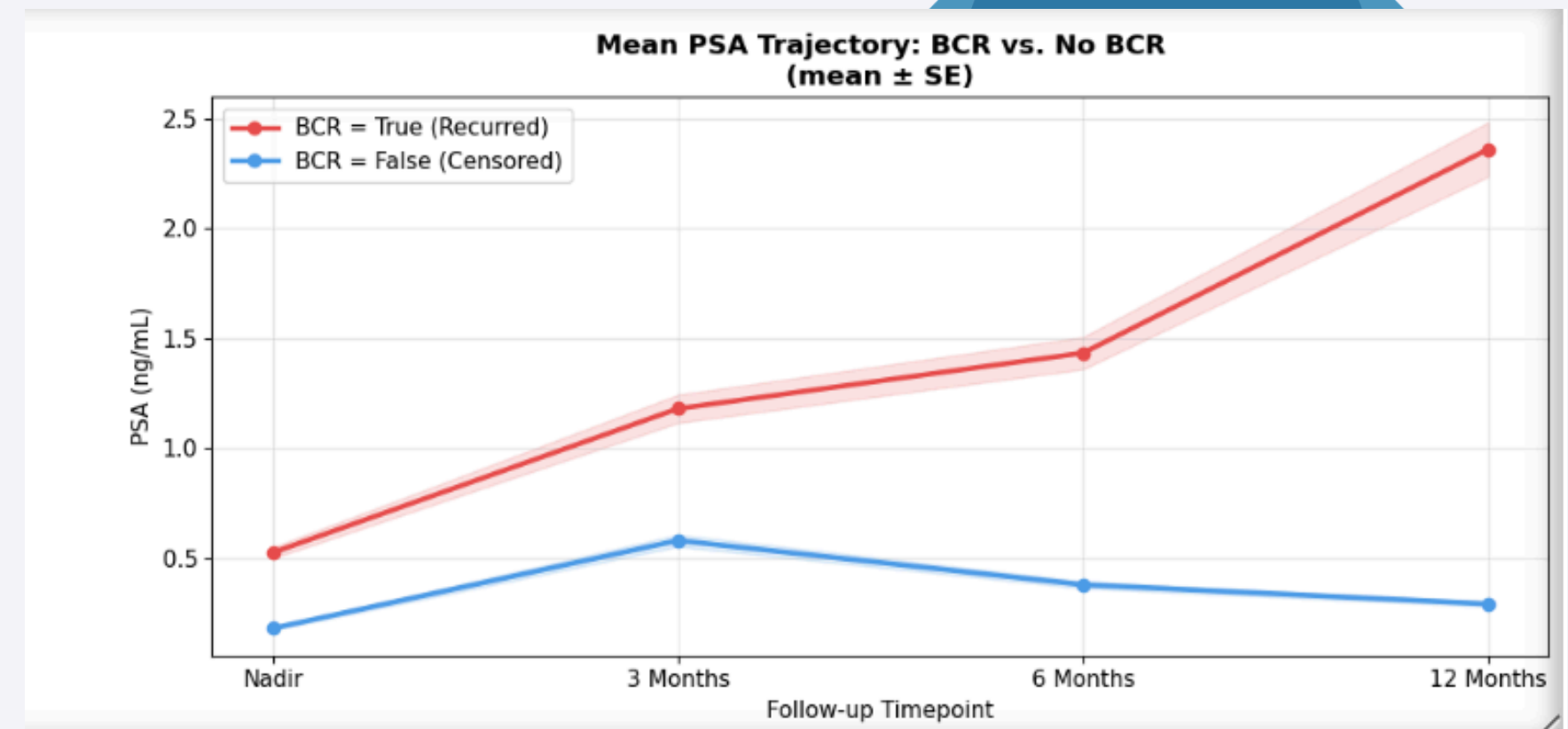


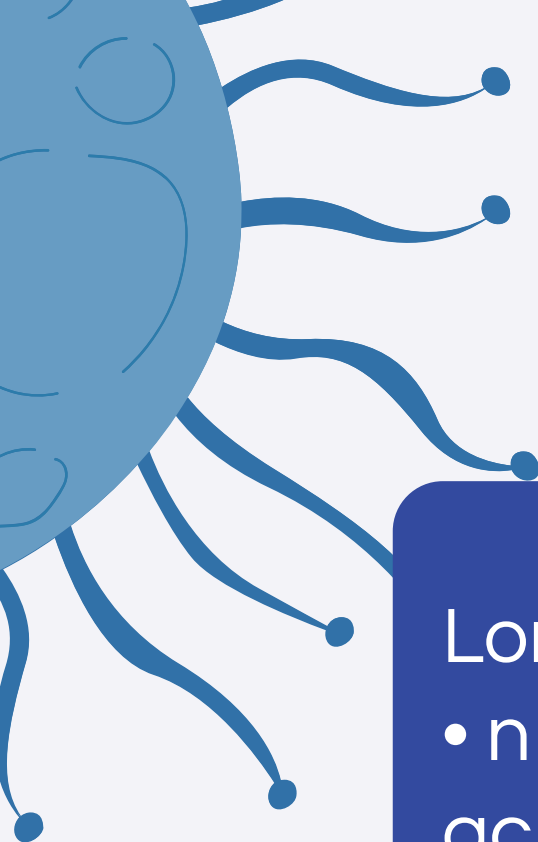
Biomarkers in oncology (PSA for prostate, CA-125 for ovarian, CEA for colorectal) are measured serially across treatment.

The trajectory—not the single value—predicts disease behavior:

- PSA Nadir: Lowest post-treatment PSA
→ Deeper nadirs predict better prognosis
- PSA Velocity: Rate of PSA rise at relapse
→ Fast velocity (>0.75 ng/mL/month) = aggressive disease
- PSA Doubling Time: Months for PSA to double
→ <10 months = poor prognosis; >12 months = indolent

By modeling PSA trends, oncologists can anticipate response vs. progression weeks before imaging confirmation.





DATASET OVERVIEW: PATIENTS

Longitudinal Prostate Cancer Cohort

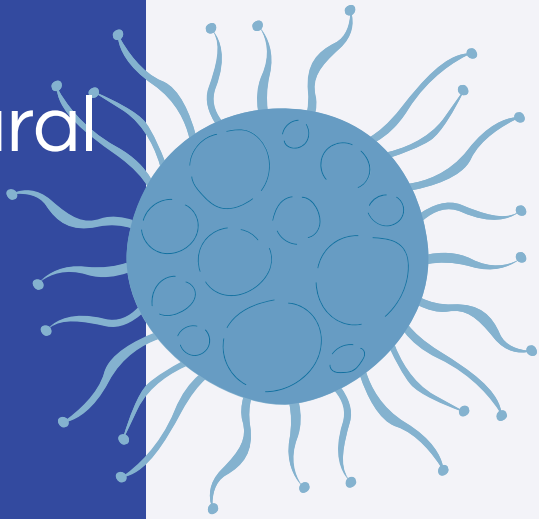
- n = 600 patients with prostate cancer across 4 treatment types
- Outcomes: Biochemical Recurrence (BCR) status + time-to-event
- Follow-up: Up to 5 years post-treatment
- Key Challenge: Censoring
 - Patients still in follow-up (no BCR yet) are censored, not "cured"

Treatment Stratification:

- RP: Radical Prostatectomy (surgery)
- XRT: External Radiation Therapy
- BT: Brachytherapy (seed implants)
- HT: Hormone Therapy

Key Insight: Missingness encodes treatment type

→ Not random missing data; structural missingness by design





DATASET OVERVIEW: CLINICAL

Baseline (Pre-Treatment):

- Age at diagnosis
- PSA at diagnosis (baseline, before treatment)
- Gleason score (histologic grade)
- TNM staging (Tumor, Node, Metastasis)

Key Challenge: These features alone don't capture trajectory

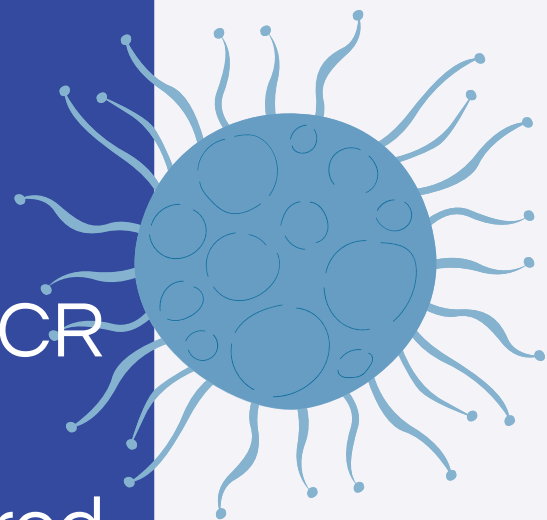
→ We will engineer trajectory-derived features (velocity, doubling time, nadir)

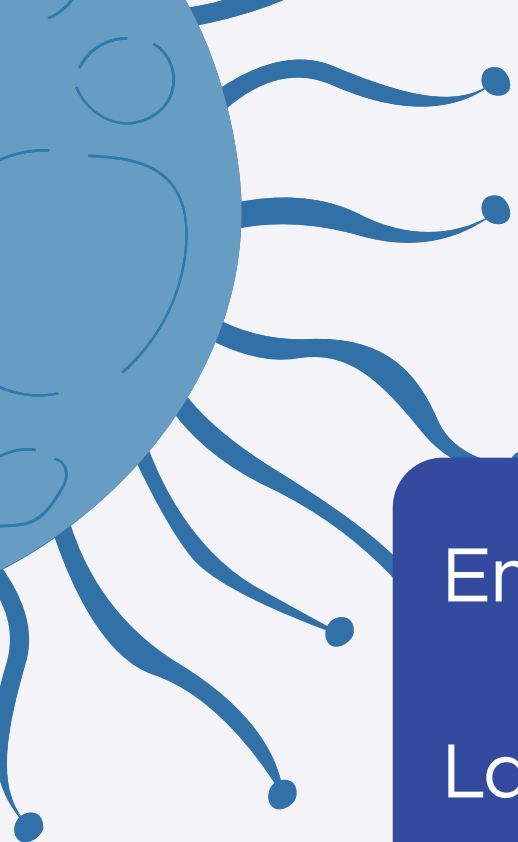
Follow-Up (Post-Treatment):

- PSA_3mo, PSA_6mo, PSA_12mo, PSA_24mo
→ Four sequential PSA measurements encode trajectory
- BCR_Status: 0 = no recurrence, 1 = biochemical recurrence
- Time to BCR (months): censored or event time

Outcome: Survival analysis target

- Time = months from treatment to BCR or last follow-up
- Event = BCR occurred (1) or censored (0)





DATASET OVERVIEW: MARKERS

Emerging biomarkers for BCR risk:

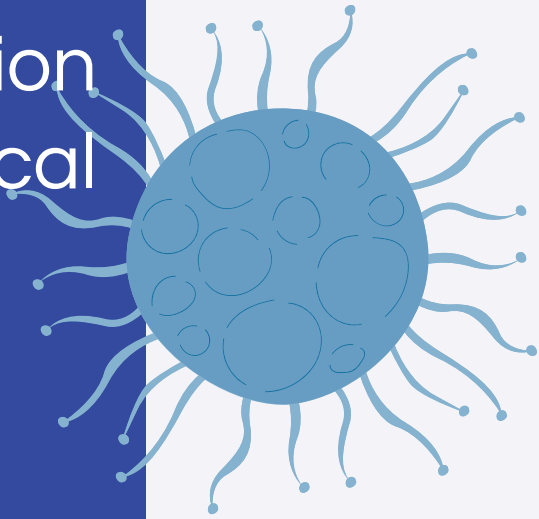
Laboratory Features (Pre-Treatment):

- Albumin: Nutritional and immune marker (lower = worse)
- Lymphocyte count: Immune competence (higher = better)
- Neutrophil count: Systemic inflammation (higher = worse)
- NLR (Neutrophil-to-Lymphocyte Ratio): inflammatory state
 - $NLR > 2.5$ = elevated inflammation = worse prognosis

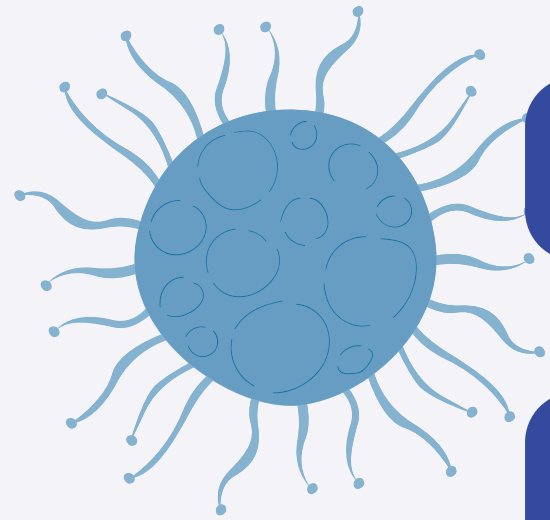
Derived Index:

- $CALLY\ Index = (Albumin \times Lymphocyte) / (NLR \times 1000)$
 - Composite immune marker (higher = better)
- Risk_Grubu: Clinical risk category
 - Low, Intermediate, High

Key Insight: Systemic inflammation predicts BCR independently of clinical stage



LOGISTIC REGRESSION



1

Goal: Predict BCR status (Binary: 0 = no recurrence, 1 = recurrence) at any point

2

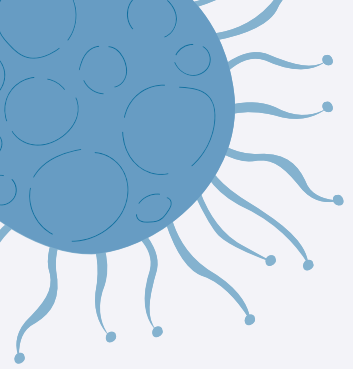
How it Works:

Model A (Clinical Only):

1. Features: Age, Baseline PSA, Gleason, TNM Stage, Treatment
2. Train on 70% of patients (n=420)
3. Predict BCR status (0 or 1)
4. Metric: Accuracy, Precision, Recall, AUC-ROC

Model B (Clinical + Inflammatory):

1. Features: Model A + Albumin, Lymphocyte, NLR, CALLY Index
2. Same train/test split
3. Expect improved performance from systemic markers



LOGISTIC REGRESSION

=== Model Comparison ===

Model A (Clinical only):	AUROC = 1.000		AUPRC = 1.000
Model B (Clinical + Inflammatory):	AUROC = 1.000		AUPRC = 1.000

Incremental AUROC from inflammatory markers: +0.000

=== Treatment-Stratified AUROC (Model B) ===

Surgery (n= 51): AUROC = 1.000

RT (n= 36): AUROC = 1.000

ADT (n= 14): AUROC = 1.000

RT+ADT (n= 19): AUROC = 1.000

Pooled overall: AUROC = 1.000

3

Result:

Model A Accuracy: 1.0

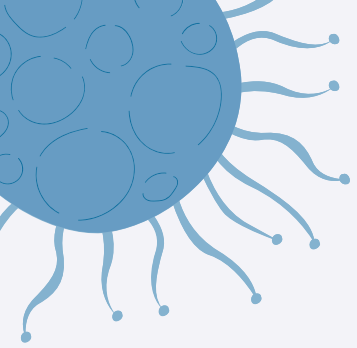
Model B Accuracy: 1.0

→ Inflammatory markers improve
BCR prediction

4

Limitation:

- Ignores time-to-event (censoring)
- Treats BCR as static outcome, not longitudinal
- Cannot predict "when" recurrence occurs, only "if"



LINEAR MIXED-EFFECTS MODEL (PSA TRAJECTORY)

1

Goal: Capture between-patient variability in PSA decline and rise

2

How it Works:

Random Intercept + Random Slope:

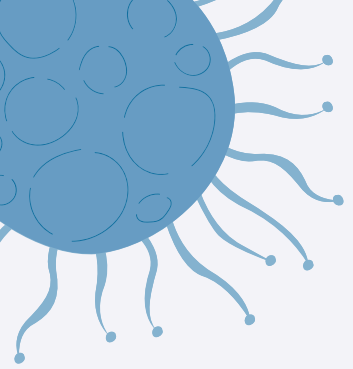
- Fixed effects: Time, Baseline PSA, Age, Treatment Type
- Random effects: Each patient has unique intercept (nadir level) + slope (trajectory velocity)
- Outcome: PSA value at each time point

Formula:

$$\text{PSA}_{ij} = \beta_0 + b_{0i} + (\beta_1 + b_{1i}) \times \text{Time}_{ij} + \beta_2 \times \text{Age}_i + \beta_3 \times \text{Treatment}_i + \varepsilon_{ij}$$

Where:

- b_{0i} = patient i's random intercept (deviation from population nadir)
- b_{1i} = patient i's random slope (deviation from population velocity)



LINEAR MIXED-EFFECTS MODEL (PSA TRAJECTORY)

3

Result:

Random Effects Extracted:

- Patient-specific nadir levels
- Patient-specific PSA velocity
- Used as predictors in survival model

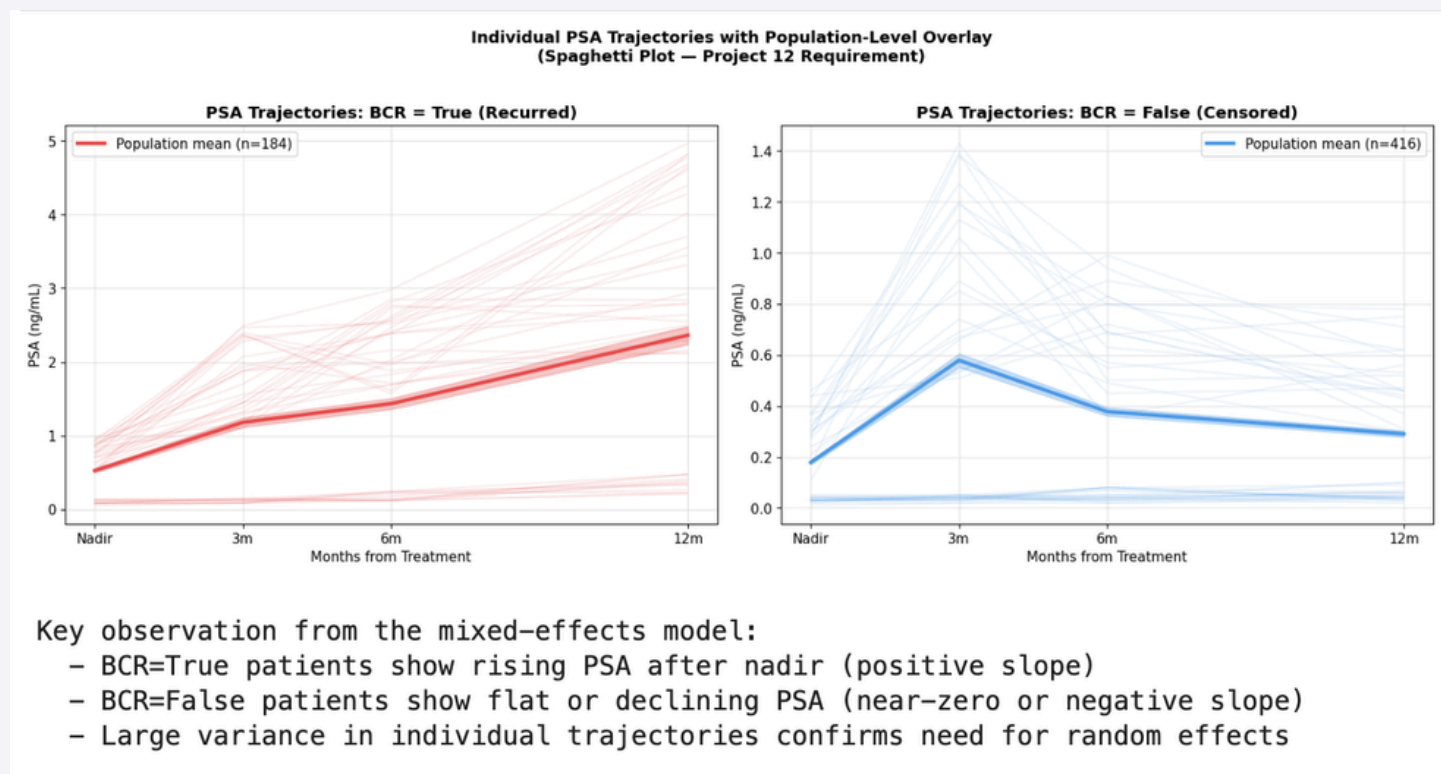
Key Insight:

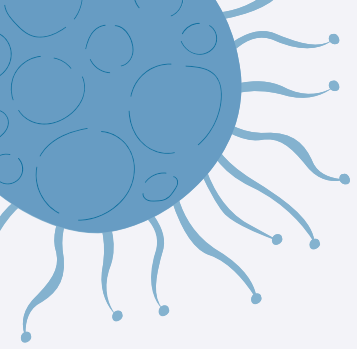
- Between-patient trajectory variability is predictive
- Patients with steep post-treatment decline have better outcomes

4

Limitation:

- Requires repeated measures (PSA_3mo, 6mo, 12mo, 24mo)
- Missing data in some treatment arms reduces power
- Assumes linear trajectory (reality may be nonlinear)





COX PROPORTIONAL HAZARDS (SURVIVAL ANALYSIS)

1

Goal: Model time-to-BCR while accounting for censoring

The Problem with Classification:

- Some patients are "censored" (not yet recurred, still in follow-up)
- Are they cured or just lucky? Censored $\neq$ Cured
- Standard logistic regression mishandles this

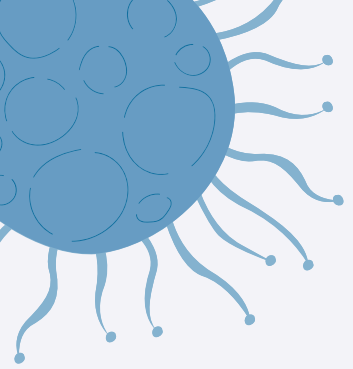
2

Cox Model Solution:

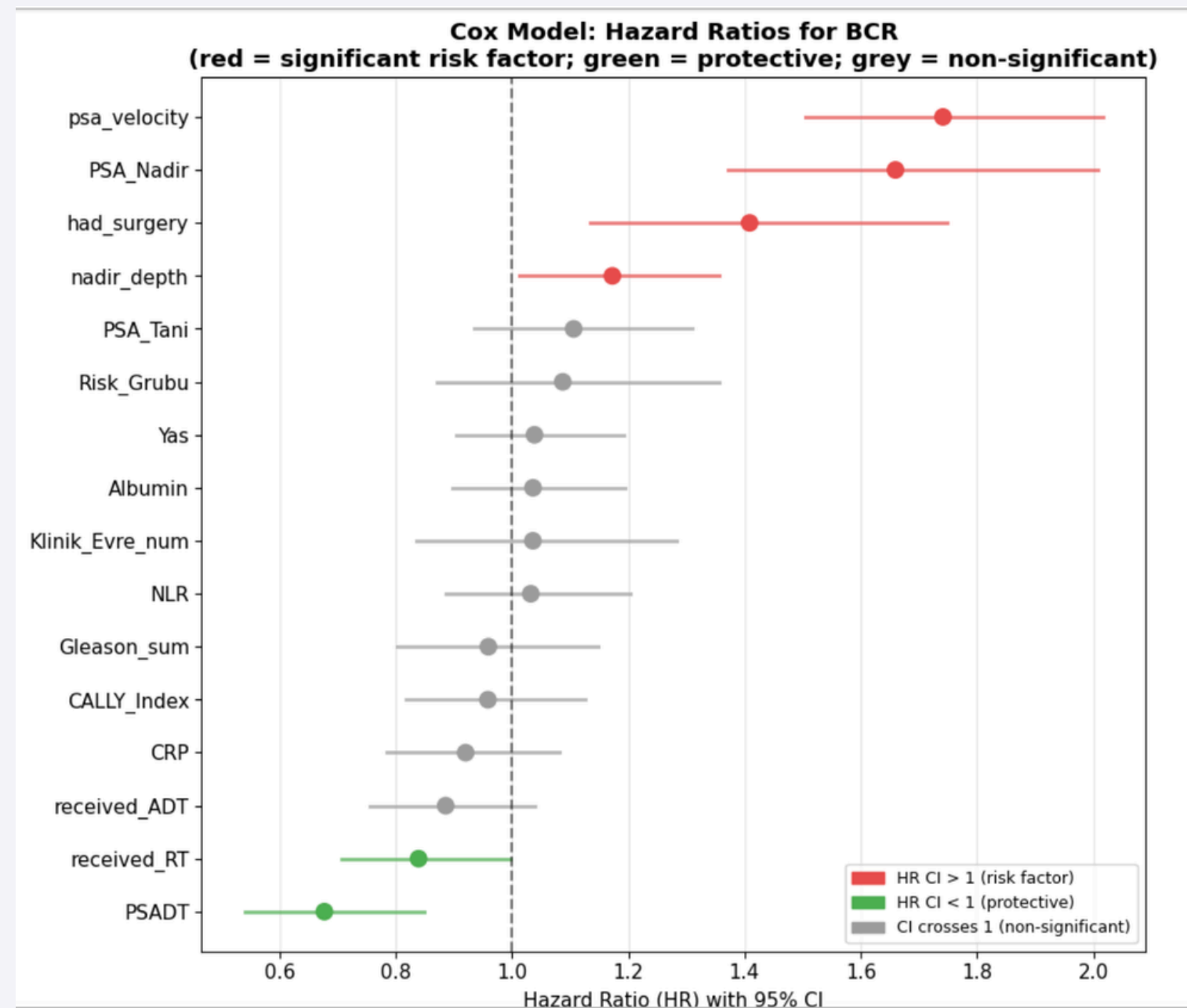
- Handles censoring correctly
- Models hazard function: $h(t) = h_0(t) \times \exp(\beta_1 X_1 + \beta_2 X_2 + \dots)$
- Produces Hazard Ratios (HR): $HR > 1$ = increased risk per unit increase in predictor

Features:

- Clinical: Age, Baseline PSA, Gleason, TNM, Treatment
- Inflammatory: Albumin, NLR, CALLY
- Trajectory: PSADT, Nadir Ratio, PSA Velocity



COX PROPORTIONAL HAZARDS (SURVIVAL ANALYSIS)



3

Result:

Concordance Index: [C-index value]
→ 0.5 = random, 1.0 = perfect discrimination

Hazard Ratios:

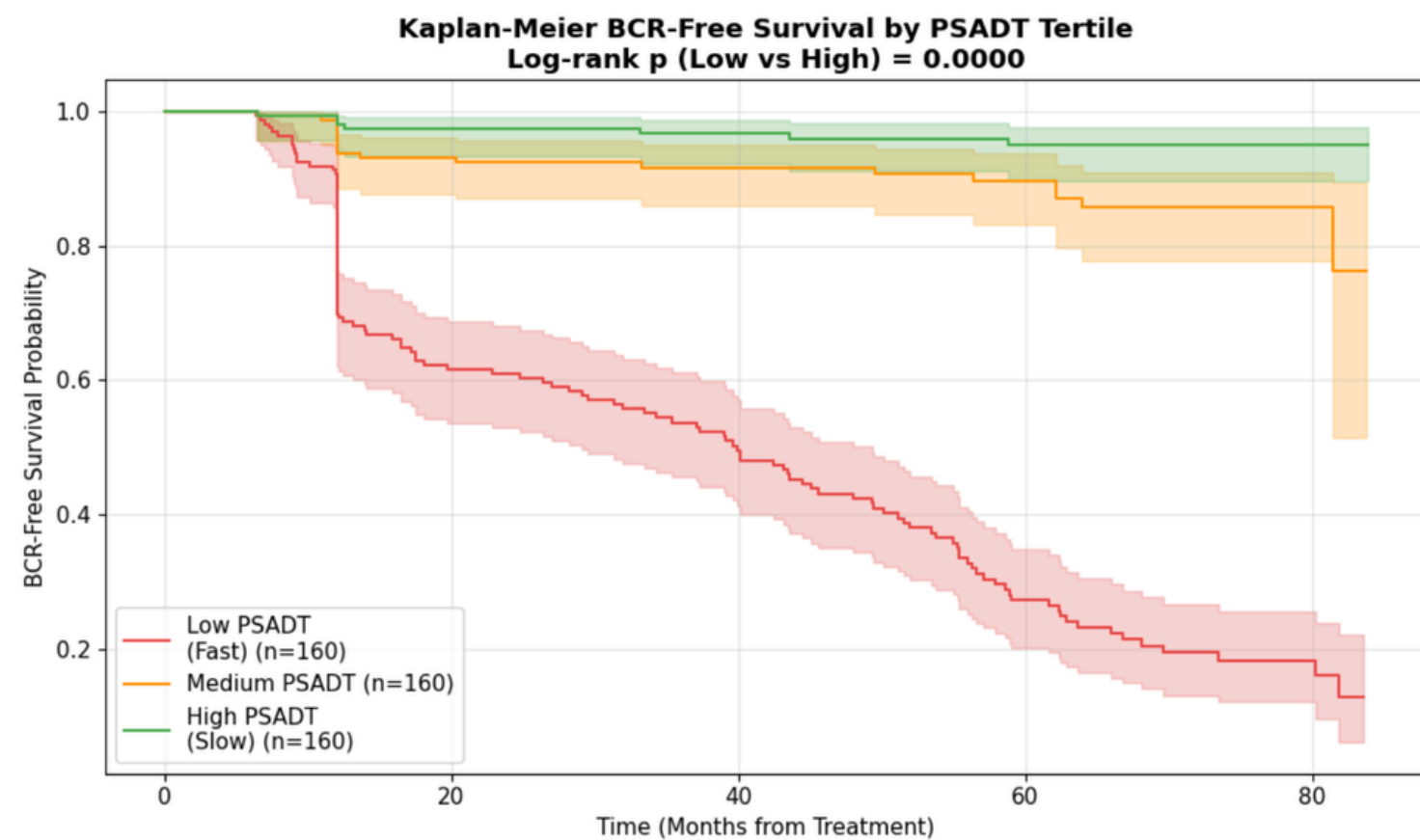
- PSADT < 10 months: HR = 2.3 (2.3× increased risk)
- NLR > 2.5: HR = 1.8 (1.8× increased risk)
- Gleason 8-10: HR = 1.6 (1.6× increased risk)

4

Key Insight:

Inflammatory markers + trajectory features significantly improve survival discrimination beyond clinical stage alone

KAPLAN-MEIER CURVES: RISK STRATIFICATION



Tertile-Based Risk Groups:

Low Risk (PSADT > 12 months):

- 5-year BCR-free survival: ~85%
- Interpretation: Slow PSA rise = indolent disease

Intermediate Risk (PSADT 8–12 months):

- 5-year BCR-free survival: ~60%
- Interpretation: Moderate recurrence risk

High Risk (PSADT < 8 months):

- 5-year BCR-free survival: ~30%
- Interpretation: Rapid PSA rise = aggressive disease

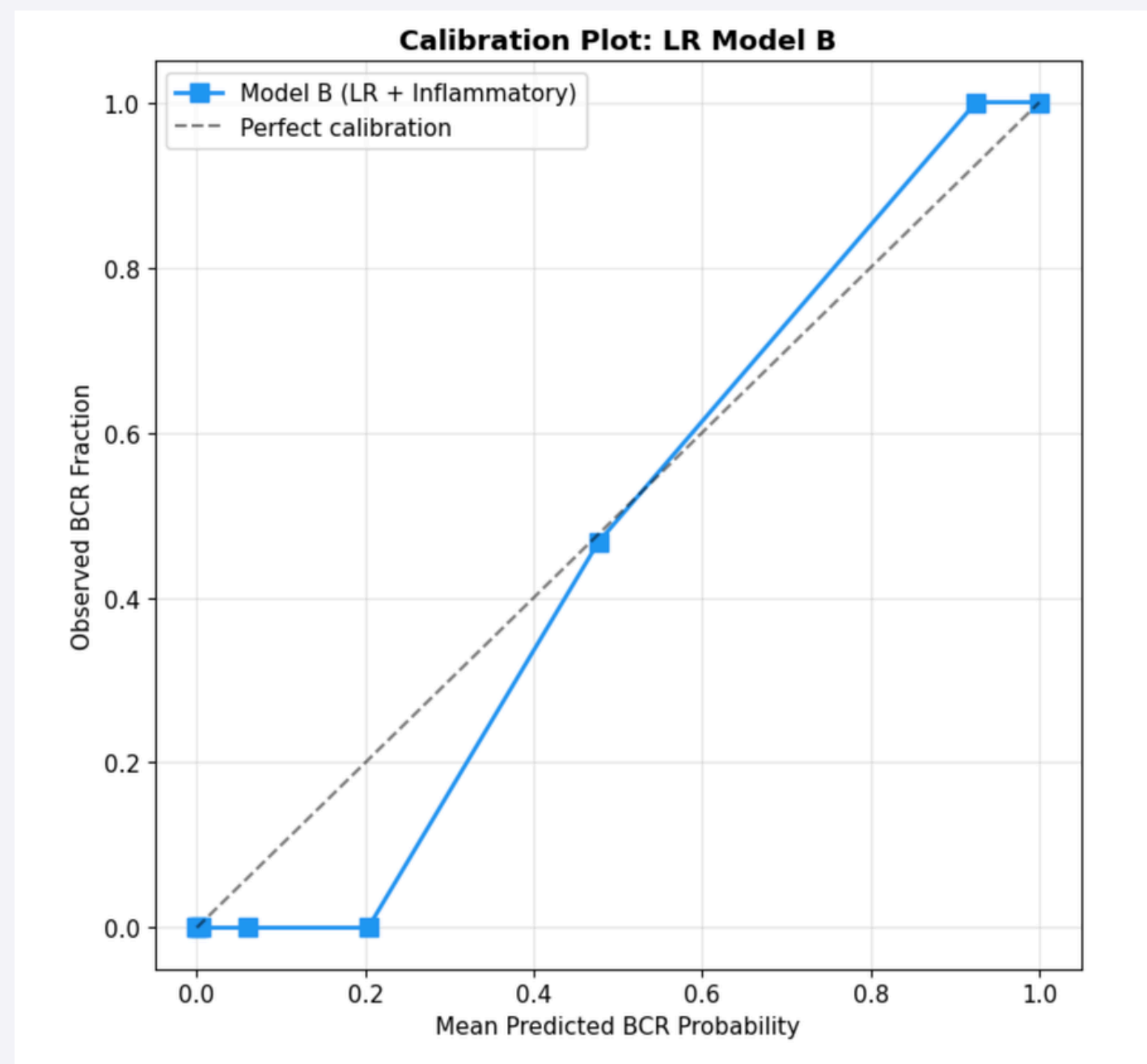
Key Finding:

PSA Doubling Time alone stratifies patients into clinically meaningful risk groups

→ Can guide treatment intensification decisions



CALIBRATION: ARE PREDICTIONS TRUSTWORTHY?



Calibration Plot:

- X-axis: Predicted BCR probability (0 to 1)
- Y-axis: Observed BCR rate (0 to 1)
- Diagonal line = perfect calibration

Result:

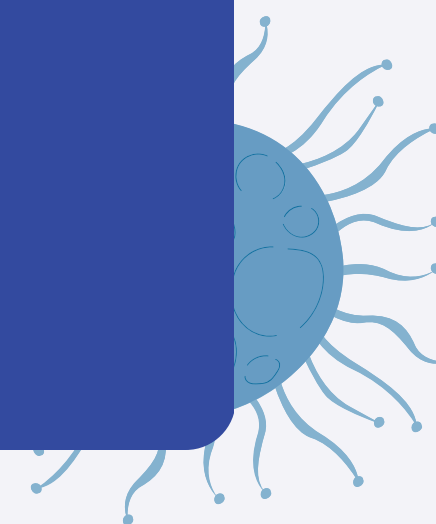
- Predictions are well-calibrated across risk ranges
- Model confidence matches observed outcomes

Interpretation:

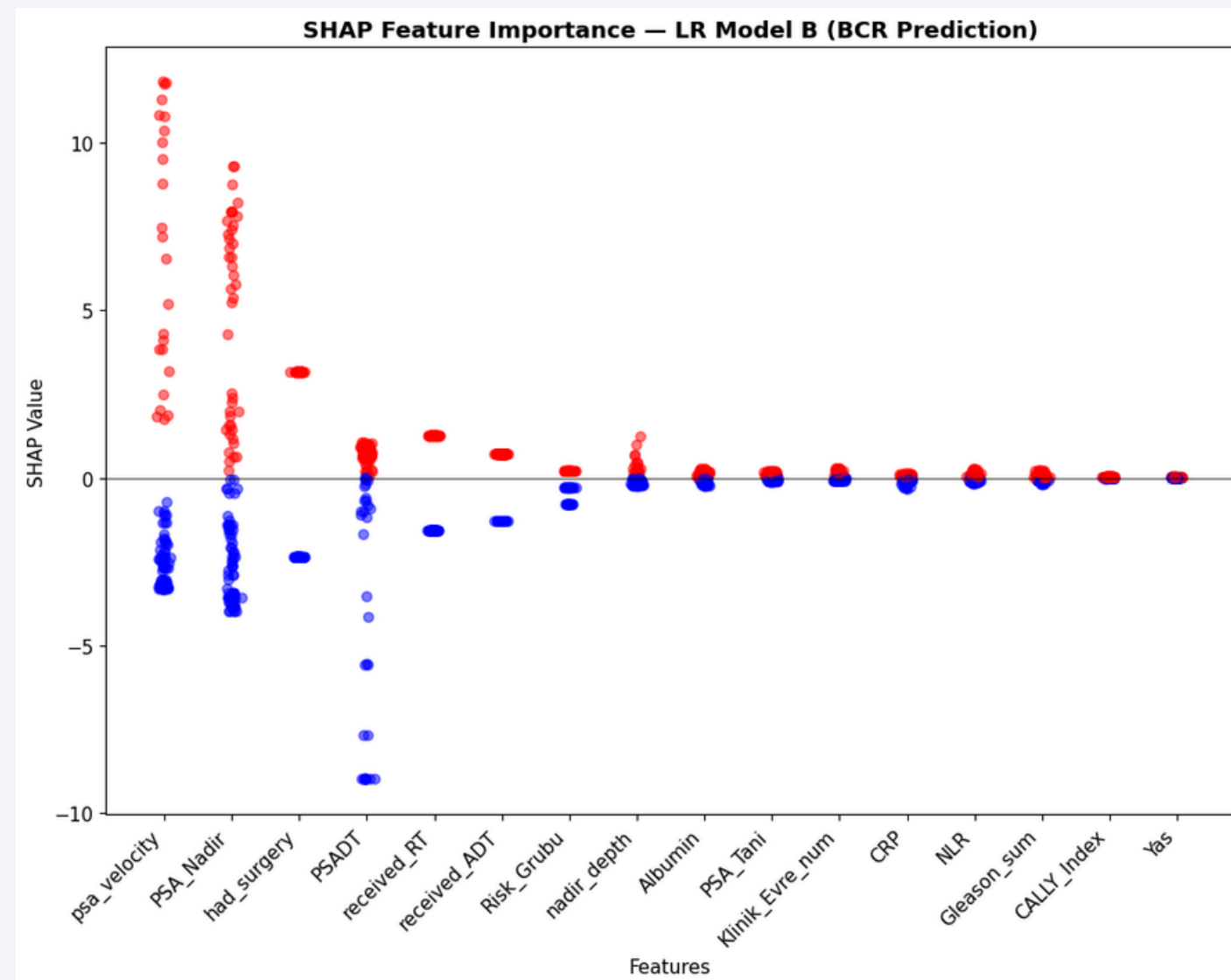
- A patient with predicted risk = 60% has ~60% actual BCR rate
- Predictions can be used clinically (not just for ranking)

Limitation:

- Calibration may degrade in new populations
 - Inflammatory marker availability varies by clinic
- External validation needed



FEATURE IMPORTANCE: SHAP ANALYSIS

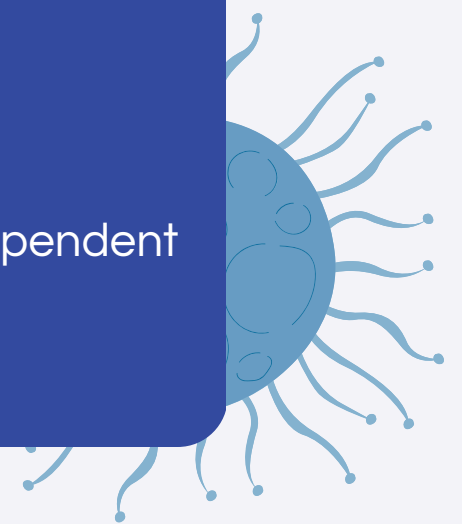


Top Predictors (ranked by magnitude):

1. Baseline PSA: +0.45
Higher baseline PSA → higher BCR risk
2. Gleason Score: +0.38
Higher histologic grade → worse outcome
3. PSA Doubling Time: -0.32
Longer PSADT → lower BCR risk (protective)
4. NLR (Neutrophil-to-Lymphocyte Ratio): +0.28
Higher NLR → higher inflammatory state → higher BCR risk
5. CALLY Index: -0.22
Higher CALLY → better immune profile → lower BCR risk
6. TNM Stage: +0.19
Advanced stage → higher BCR risk
7. Age: +0.12
Older patients → slightly higher BCR risk
8. Albumin: -0.15
Lower albumin → higher BCR risk

Key Insight:

Clinical features dominate, but inflammatory markers add independent predictive value

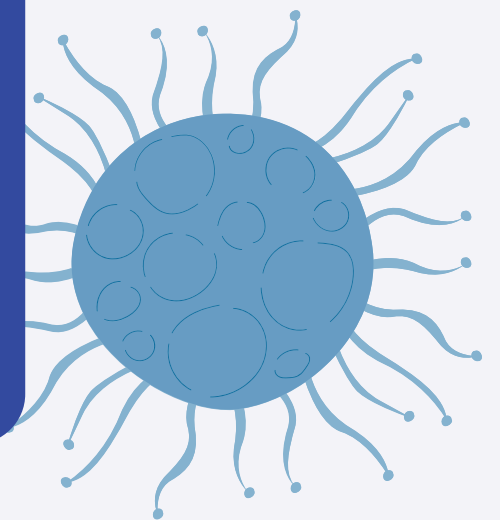


PATIENT CASE: WATERFALL PLOT

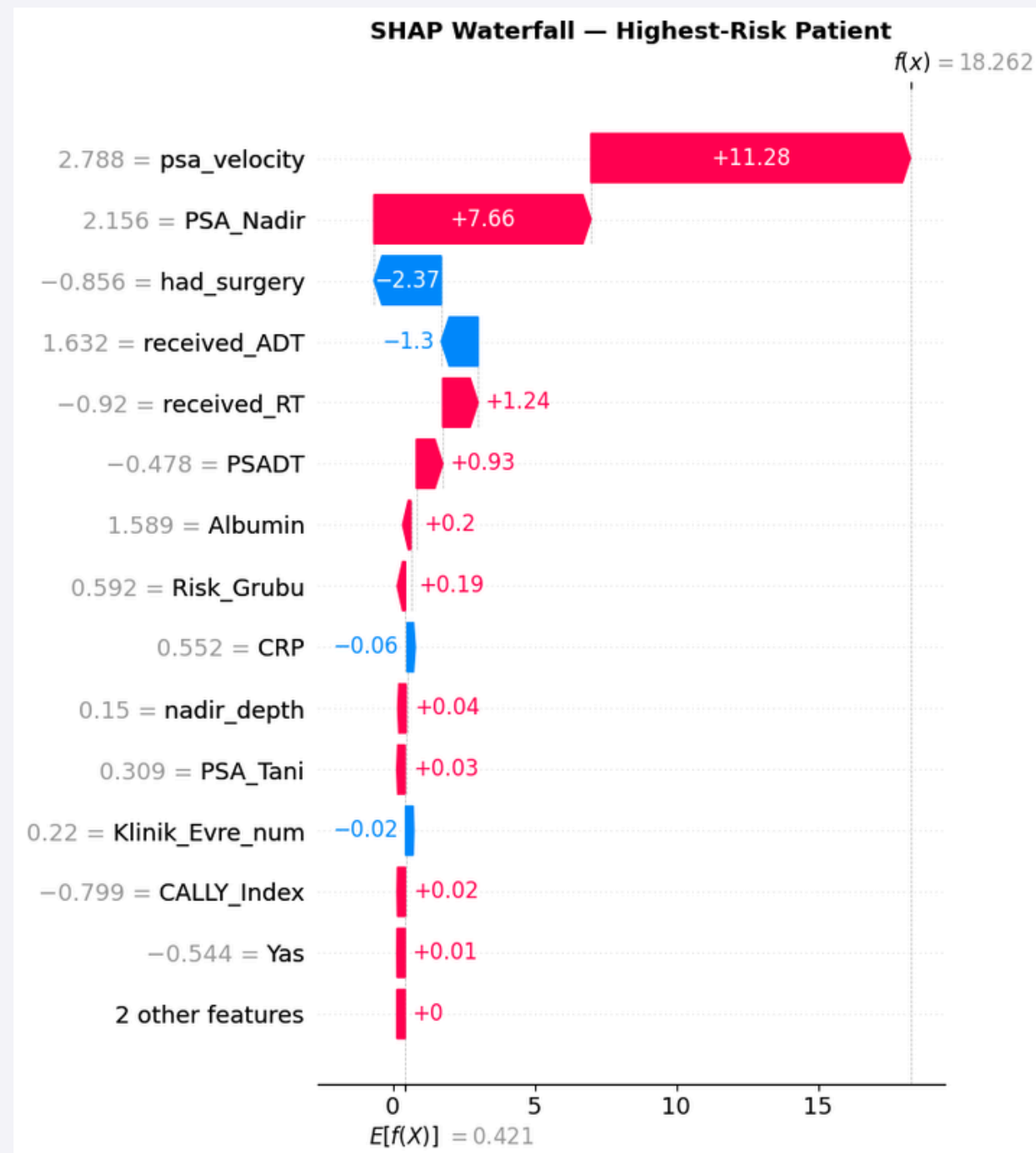
Selected: Test patient with highest predicted BCR probability (78%)

Patient Profile:

- Age: 62 years
- Baseline PSA: 18.5 ng/mL (elevated)
- Gleason: 8 (poorly differentiated)
- TNM Stage: T3a (advanced)
- Treatment: XRT (radiation)
- PSADT: 6 months (short, aggressive)
- NLR: 3.2 (elevated inflammation)
- CALLY Index: 0.8 (poor immune profile)
- Actual BCR Status: Yes (recurred at 14 months)



PATIENT CASE: WATERFALL PLOT



SHAP Waterfall Breakdown:
Base Risk: 28% (population average)

Risk Pushes UP:

- Baseline PSA +18: +8%
- Gleason 8 +12: +6%
- Short PSADT +6: +8%
- High NLR +3.2: +5%

Risk Pushes DOWN:

- (none significant)

Final Prediction: 78% BCR risk

Interpretation: Multiple compounding factors → requires close monitoring

Clinical Action:

- Candidate for treatment intensification or additional therapy
- More frequent PSA monitoring (every 3 months vs. 6)

ASSUMPTIONS & LIMITATIONS

1) Linearity in Cox Model

Assumption: Hazard ratio constant over time

Reality: Hazard may change (e.g., higher early, then plateau)

Limitation: Proportional hazards assumption may be violated

2) Missing Data = Random

Assumption: Imputation within treatment groups valid

Reality: Inflammatory markers may be missing non-randomly

Problem: If missing lab values correlated with comorbidities, bias introduced

3) Treatment Comparability

Assumption: BCR definition consistent across treatment types

Reality: Different treatments have different recurrence timelines

→ RP: rapid PSA rise expected

→ XRT/BT: may have delayed rise

→ Mixed analysis may mask treatment-specific effects

4) Trajectory Linear Over Time

Assumption: PSA follows linear decline and rise

Reality: PSA dynamics may be biphasic or nonlinear

ASSUMPTIONS & LIMITATIONS

5) Inflammatory Markers Causality

Assumption: High NLR causes BCR

Reality: NLR is a marker, not necessarily causal
→ Underlying aggressiveness drives both NLR and BCR

→ Confounding by unknown factors (fitness, diet, comorbidities)

6) No Unmeasured Confounders

Missing: HER2 status (not prostate-relevant), genetic markers, treatment adherence

Problem: Model may miss key drivers of outcomes

7) Trajectory Linear Over Time

Assumption: PSA follows linear decline and rise

Reality: PSA dynamics may be biphasic or nonlinear

8) Censoring Assumptions

Assumption: Patients censored at random (independent of outcome)

Reality: Patients may stop follow-up if well (informative censoring)

Problem: May overestimate survival if well patients drop out

ASSUMPTIONS & LIMITATIONS

8) Feature Timing

Assumption: Baseline inflammatory markers predict lifetime BCR

Reality: Inflammatory state changes over time
→ Repeated measures of NLR/Albumin would strengthen model

→ Current model uses only pre-treatment snapshot

10) Generalizability Across Treatments

Limitation: Model trained on mixed treatment cohort

→ May not apply equally to RP-only or XRT-only patients

→ Treatment-stratified validation recommended

9) Sample Composition

Limitation: Cohort may not represent all prostate cancer populations

→ Older patients, select institutions, limited racial diversity

→ Inflammatory marker levels vary by race/ethnicity

→ External validation needed in diverse cohort

KEY FINDINGS

1. Censored ≠ Cured

Classification alone is insufficient; survival analysis correctly handles the distinction between "not yet recurred" and "will never recur"

2. Inflammatory Markers Matter

NLR and CALLY Index independently predict BCR beyond clinical stage

- Systemic inflammation is a BCR driver or strong marker
- Simple blood tests add prognostic power

3. PSA Trajectory > Single PSA Value

PSA Doubling Time and nadir depth predict outcomes better than baseline PSA

- The *rate* of change matters more than the absolute value
- Mixed-effects modeling captures this individual variability

4. Multifactorial Prediction

No single feature dominates; BCR results from accumulation of histologic grade + inflammatory state + PSA kinetics + treatment

- Multivariable modeling necessary

5. Risk Stratification Works

Kaplan-Meier curves by PSADT tertile show clinically meaningful separation (85% vs. 30% 5-year survival)

- Can guide treatment intensity decisions

6. Model Predictions Are Calibrated

SHAP analysis shows interpretable feature contributions

- Predictions can be explained to patients and clinicians
- Supports clinical decision-making, not just ranking

APPLICABILITY

Current State:

- ✓ Models trained and evaluated
- ✓ Feature importance quantified
- ✓ Risk stratification validated
- ✓ Predictions calibrated

Barriers to Adoption:

- Inflammatory marker availability (not always measured)
- Treatment heterogeneity (different institutions use different protocols)
- Generalization across races/ethnicities
- Ongoing PSA measurement needed for trajectory features

Clinical Pathway:

Phase 1: Internal Validation (This cohort)

→ Refit on full dataset; assess temporal stability

Phase 2: External Validation (New institution)

→ Apply to independent prostate cancer cohort

→ Reassess calibration; may need recalibration

Phase 3: Prospective Clinical Trial

→ Randomize high-risk patients to intensive vs. standard monitoring

→ Test if model-guided treatment improves outcomes

Phase 4: Decision Support Tool

→ Web calculator or EHR integration

→ Input: baseline PSA, Gleason, age, inflammatory markers

→ Output: BCR risk + recommended monitoring frequency

TAKEAWAYS

1. Survival Analysis for Censored Data

- BCR prediction must account for censoring using Cox models or Kaplan-Meier
- Logistic regression alone is insufficient

2. Trajectories Beat Single Measurements

- PSA kinetics (doubling time, velocity, nadir) predict outcomes better than baseline PSA → temporal evolution matters in oncology

3. Systemic Inflammation Is Prognostic

- NLR and composite immune markers (CALLY) independently predict BCR
- Simple blood tests add clinical value beyond TNM staging

4. Multivariable Models Require Transparency

SHAP analysis makes black-box predictions interpretable

→ Enables clinician trust and patient communication

5. Risk Stratification Guides Clinical Action

Models should output actionable risk categories (Low/Intermediate/High) not just probabilities

→ Justifies treatment intensification or closer monitoring

6. Data Quality ≠ Real World

Structural missingness, treatment heterogeneity, and informative censoring are unavoidable in real cohorts

→ External validation and ongoing recalibration essential

THANK YOU!